Supplementary Table S1. Validation-selected confidence thresholds, retained checkpoint metadata, and classical parameter and box-construction settings for the 23 methods.

Supplementary Table S2. Deep-learning training summaries, including the epoch and mAP50–95 at the validation-curve maximum and the number of epochs recorded.

Supplementary Table S3. Average precision at IoU thresholds from 0.50 to 0.95 for all 23 methods.

Supplementary Table S4. Centroid localization metrics for all 23 methods, including centroid precision, recall, F1 and localization-error summaries at matching radii from 2 to 20 pixels in 2-pixel increments.

Supplementary Table S5. CPU and GPU inference-time and throughput measurements for the evaluated methods.